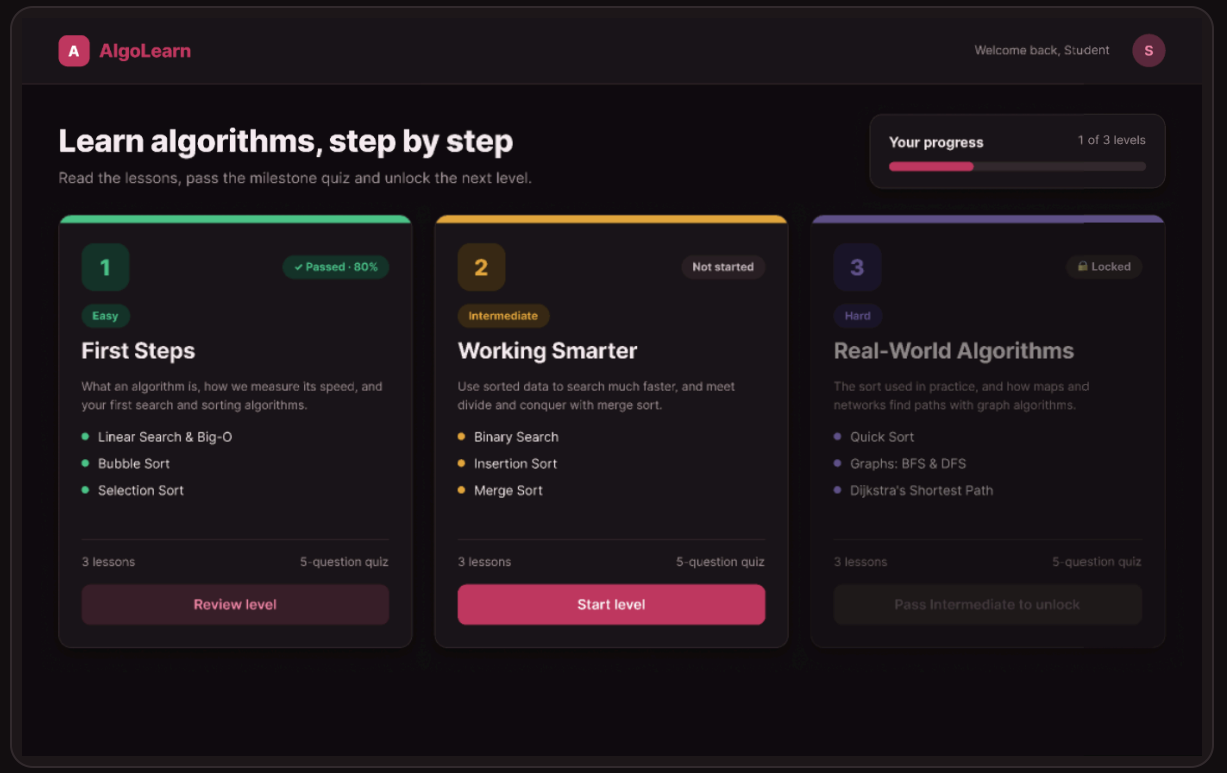
A

AlgoLearn

Short lessons, live animations and a quiz that unlocks the next level.

First presentation · Design & team



SD Squad

Toumani Adib · Alhamad Solaiman · Youssef Jelassi · Feryel

The idea

Algorithms are hard to learn from text alone

AlgoLearn is a desktop app where students learn classic algorithms one level at a time. Every lesson explains the idea in plain words, shows real Java code, and lets you watch the algorithm run.

1

Read the lesson

A short explanation, the steps, Java code and the time complexity.

2

Watch it run

An animation shows every comparison and swap, step by step.

3

Take the quiz

5 questions at the end of the level, with feedback on every answer.

4

Level up

Score 60% or more and the next level opens.

Content

Three levels, from first steps to real-world algorithms

1

Easy

First Steps

- Linear Search & Big-O
- Bubble Sort
- Selection Sort

3 lessons · 5-question quiz

2

Intermediate

Working Smarter

- Binary Search
- Insertion Sort
- Merge Sort

3 lessons · 5-question quiz

3

Hard

Real-World Algorithms

- Quick Sort
- Graphs: BFS & DFS
- Dijkstra's Shortest Path

3 lessons · 5-question quiz

Pass mark: 60%. Get at least 3 of 5 questions right in a level's milestone quiz to unlock the next level.

Team

Who does what in Phase 1

T

Toumani Adib

UI/UX design

Designed every screen, the design system and the clickable prototype in Figma.

S

Alhamad Solaiman

Learning material

Wrote the 9 lessons: explanation, steps, Java code and complexity.

Y

Youssef Jelassi

Quiz questions

Wrote the 15 milestone quiz questions with answers and explanations.

F

Feryel

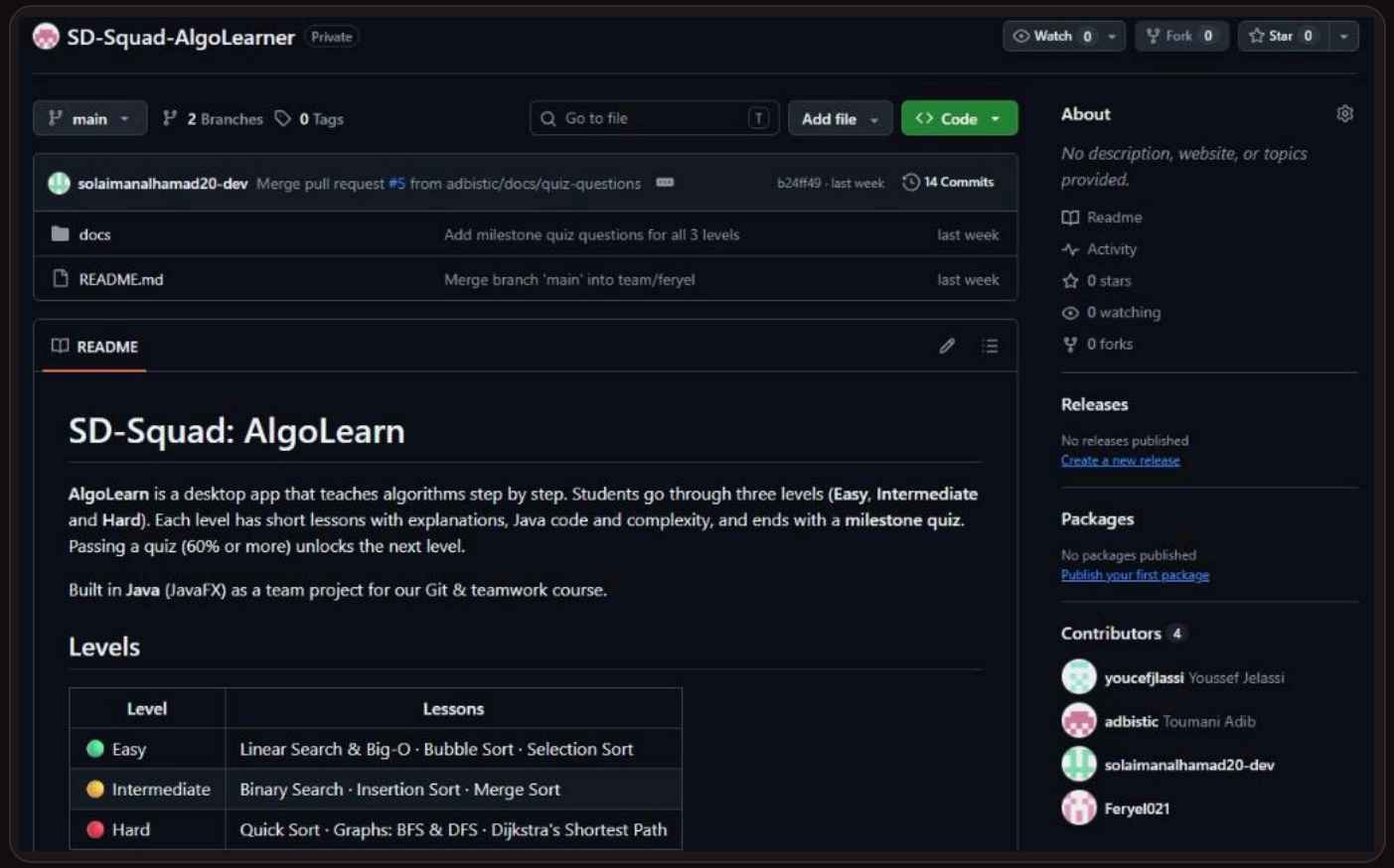
Presentation

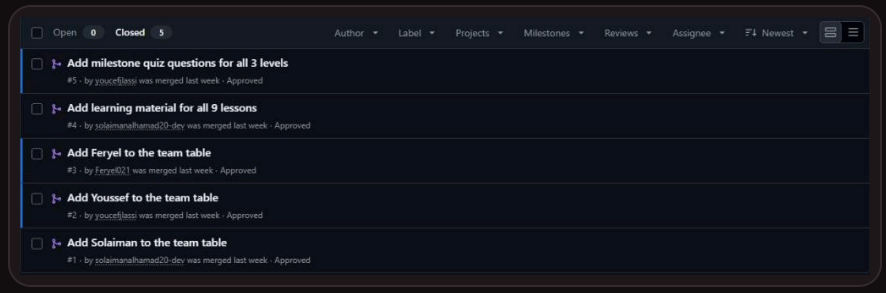
Built this presentation: the idea, the team, our workflow and the design.

Everyone commits to the same GitHub repository and keeps their own cards on Trello.

Workflow

Our GitHub repository





Every pull request was reviewed and approved before merging.

4 **contributors**

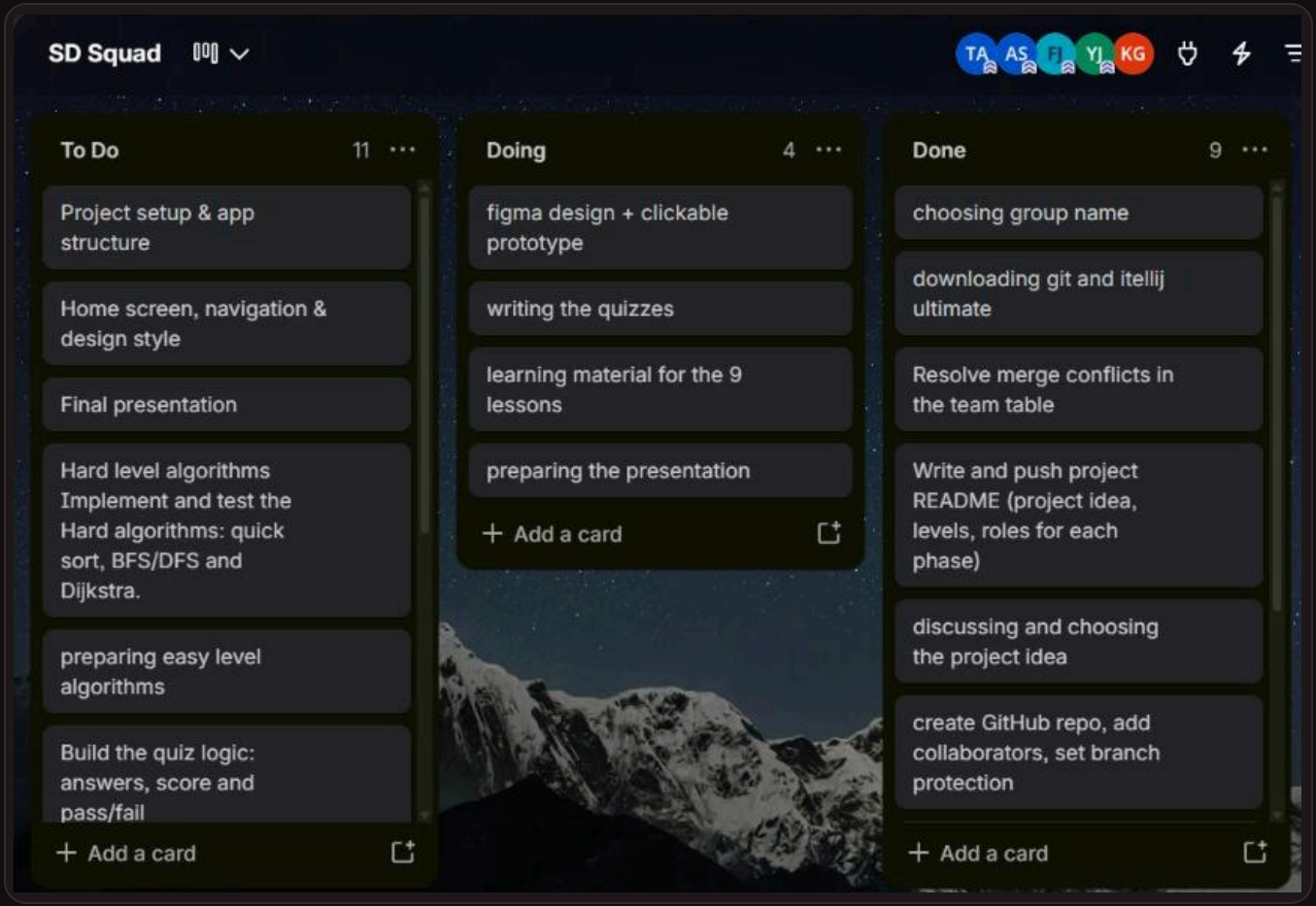
Toumani, Solaiman, Youssef and Feryel all commit to the same repository.

How every change gets in

Branch › **Pull request** › **Review** › **Merge**

Workflow

Our Trello board



Every member creates their own cards and moves them from To Do to Done as they work.

To Do
Tasks planned but not started yet

Doing
What each member is working on right now

Done
Finished tasks, kept as a record of our work

Figma

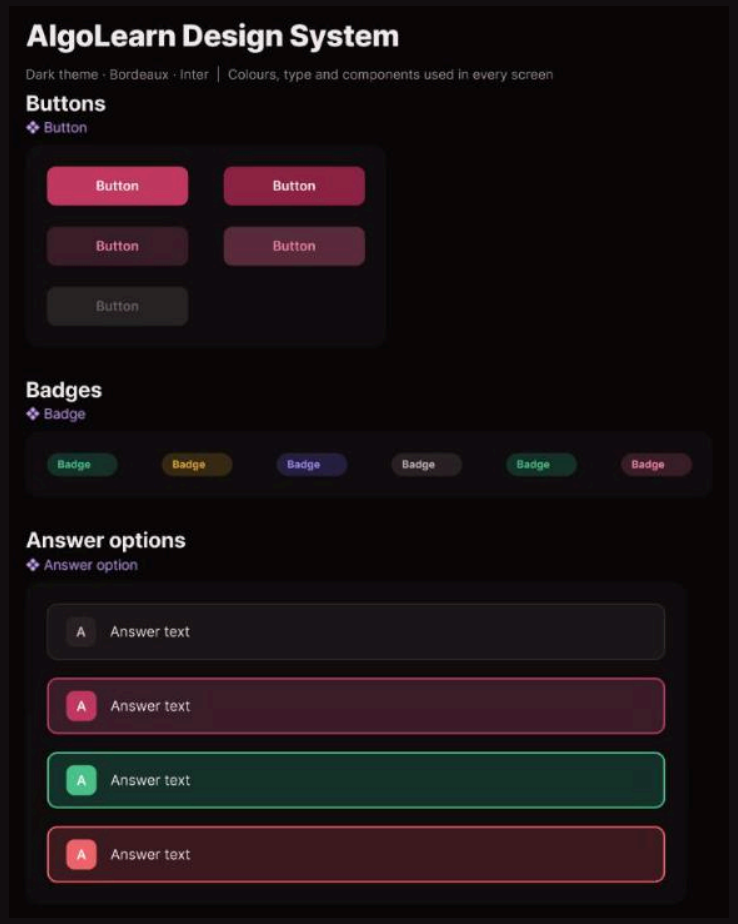
From idea to clickable prototype

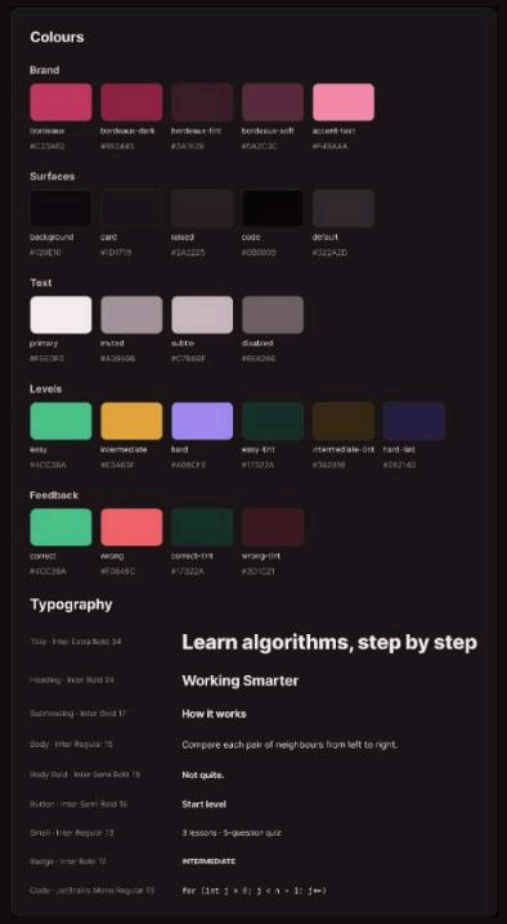
16 screens, one design system and a prototype you can click through like the real app.



Design system

One design system behind every screen





Colours

One bordeaux brand colour on dark surfaces, plus a colour for each level: green, amber and purple.

Type

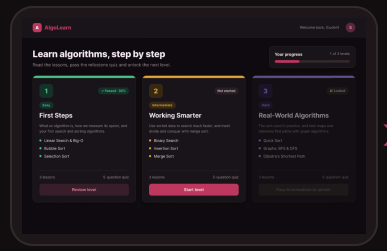
One font (Inter) in 9 text styles, from 34 pt titles down to 13 pt code.

Components

Buttons, badges and answer options with states (hover, selected, correct, wrong), reused on every screen.

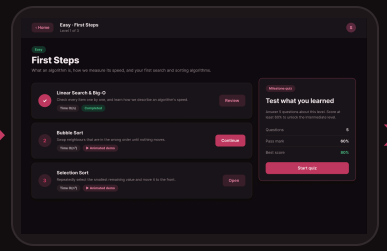
User flow

How a student moves through the app



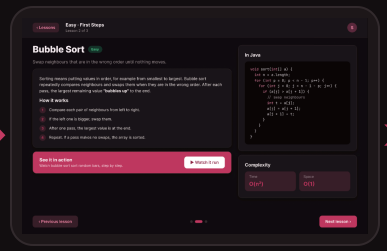
Home

Pick a level



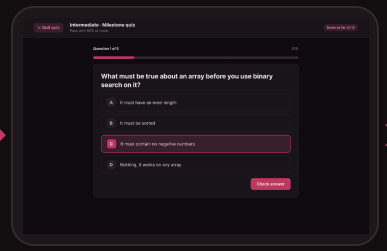
Level

See the 3 lessons



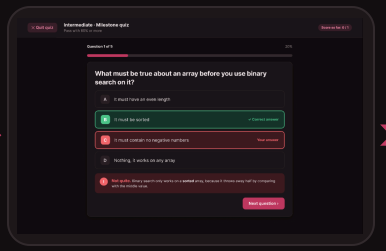
Lesson

Read and watch



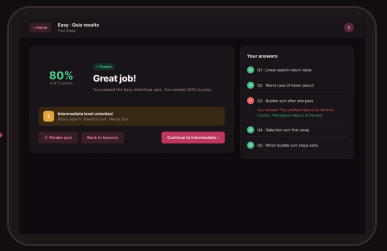
Quiz

Answer 5 questions



Feedback

Learn from mistakes



Results

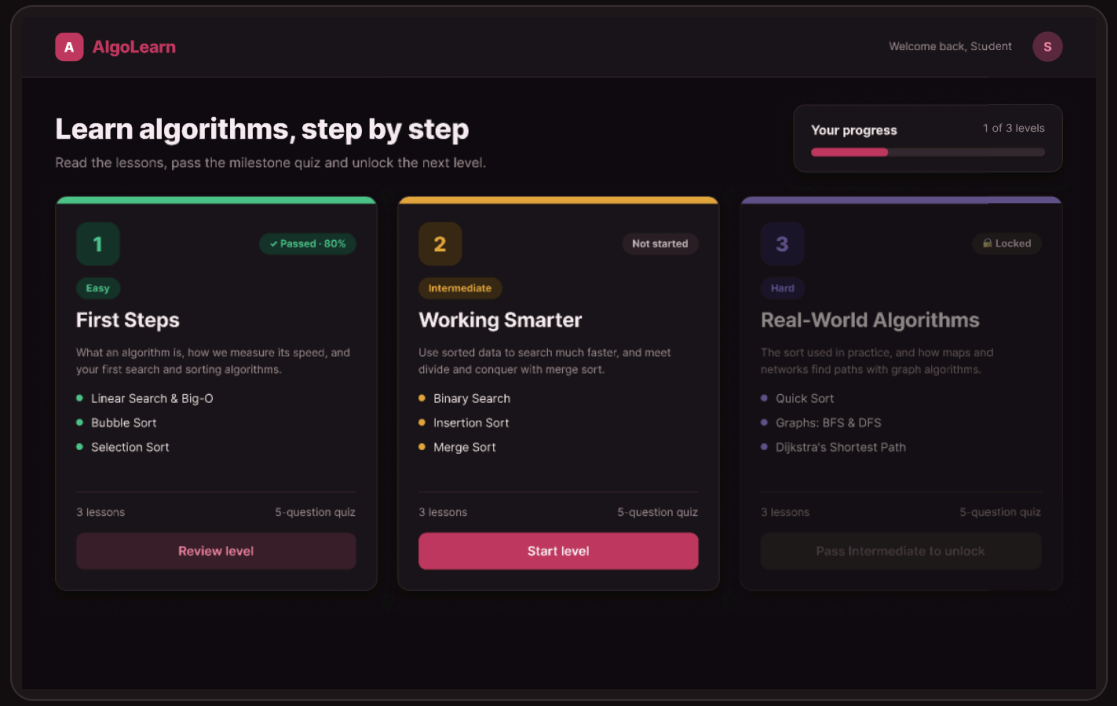
Next level unlocked

From any lesson “Watch it run” opens the sorting visualizer.

Below 60% retake the quiz or go back to the lessons.

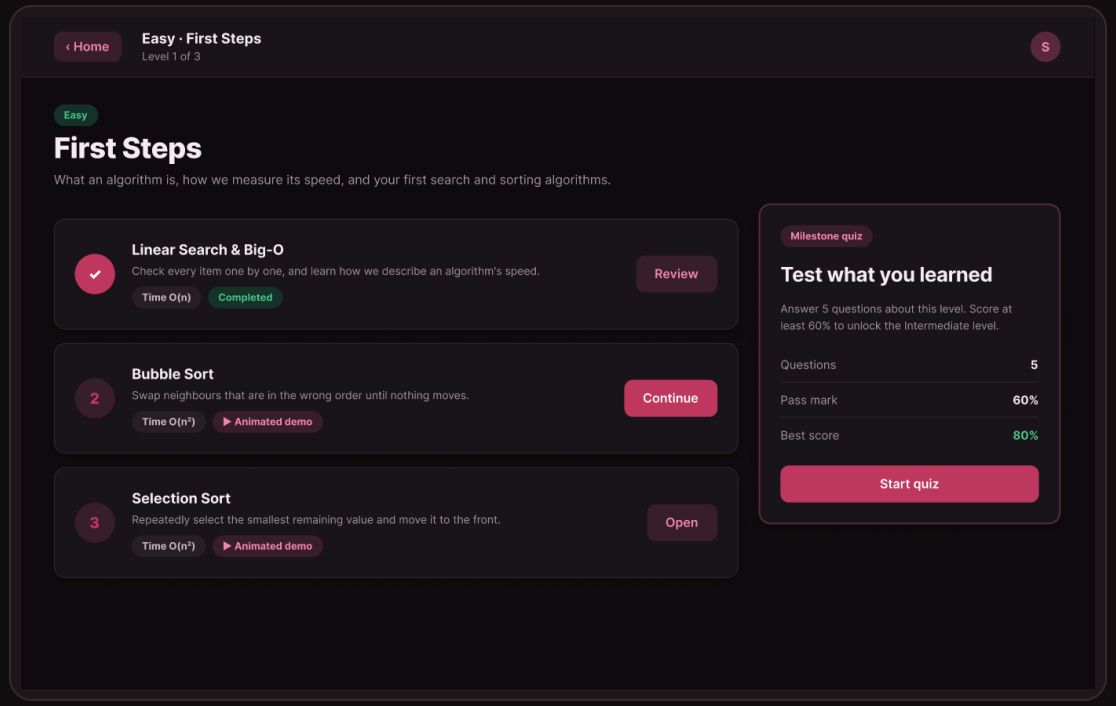
Screens

Home and level pages



Home

All three levels and your progress. Hard stays locked until Intermediate is passed.

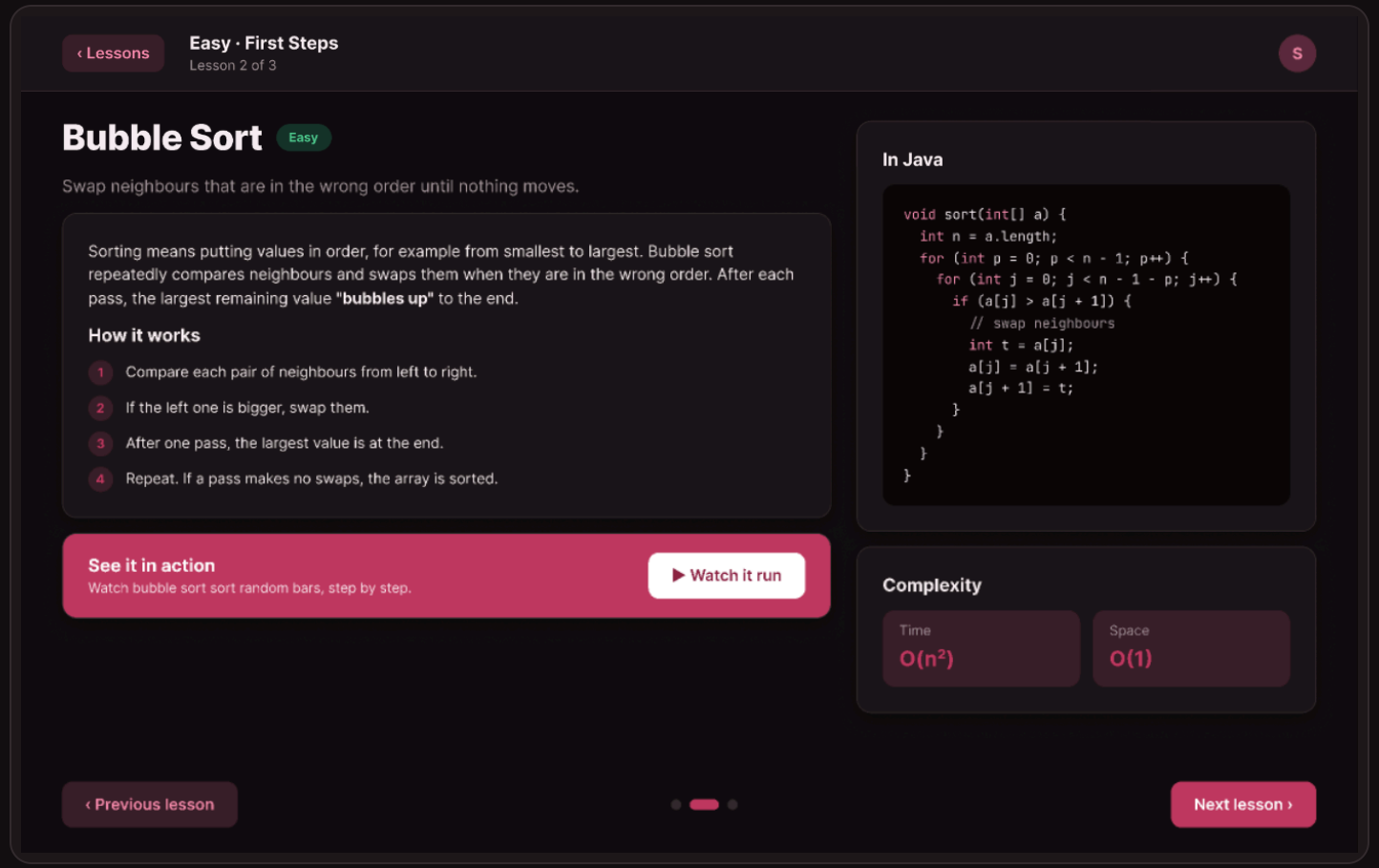


Level page

The three lessons with their status, plus the milestone quiz card.

Screens

Lessons you can read and watch

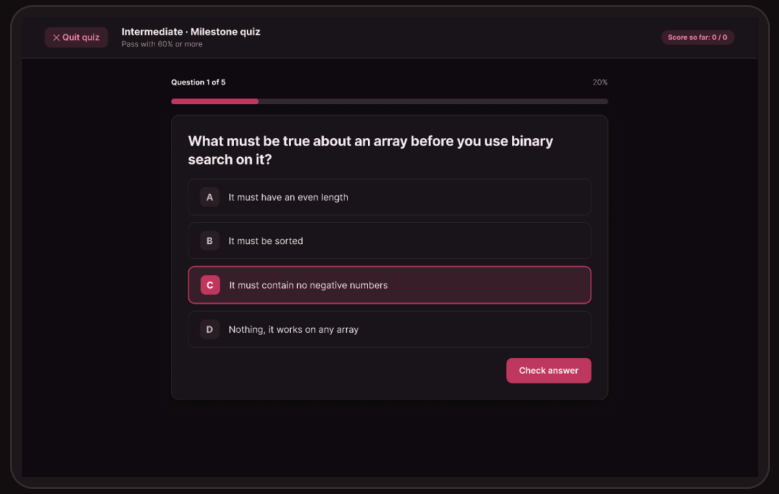


- **Plain explanation**
  What the algorithm does, in simple words and numbered steps.
- **Real Java code**
  The same code students will write, with highlighting.
- **Complexity**
  Time and space in Big-O, for every algorithm.



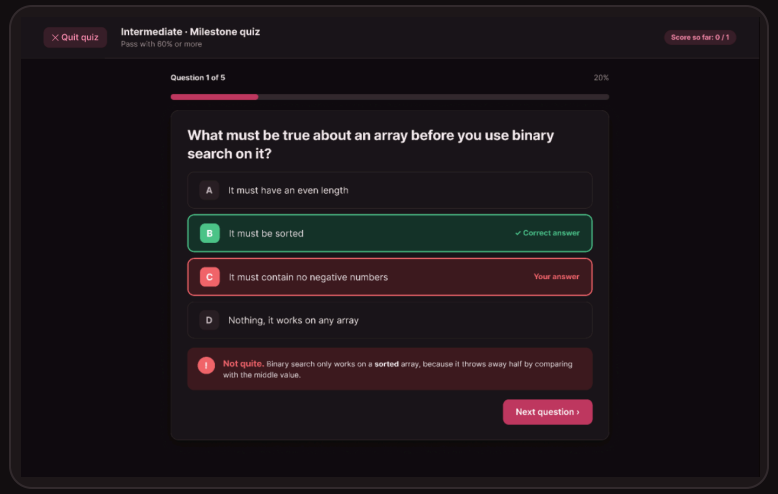
Screens

Milestone quiz with instant feedback



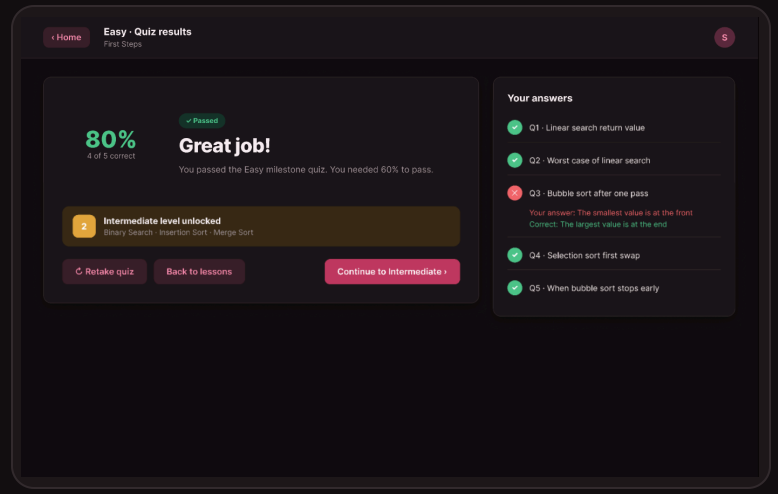
1 Answer

Pick one of four options and press Check answer.



2 Feedback

The right answer turns green, a wrong one red, with a short explanation.



3 Results

Your score, which answers were wrong, and the next level unlocked.

15 questions in total 5 per level, each with an explanation, so a wrong answer still teaches something.

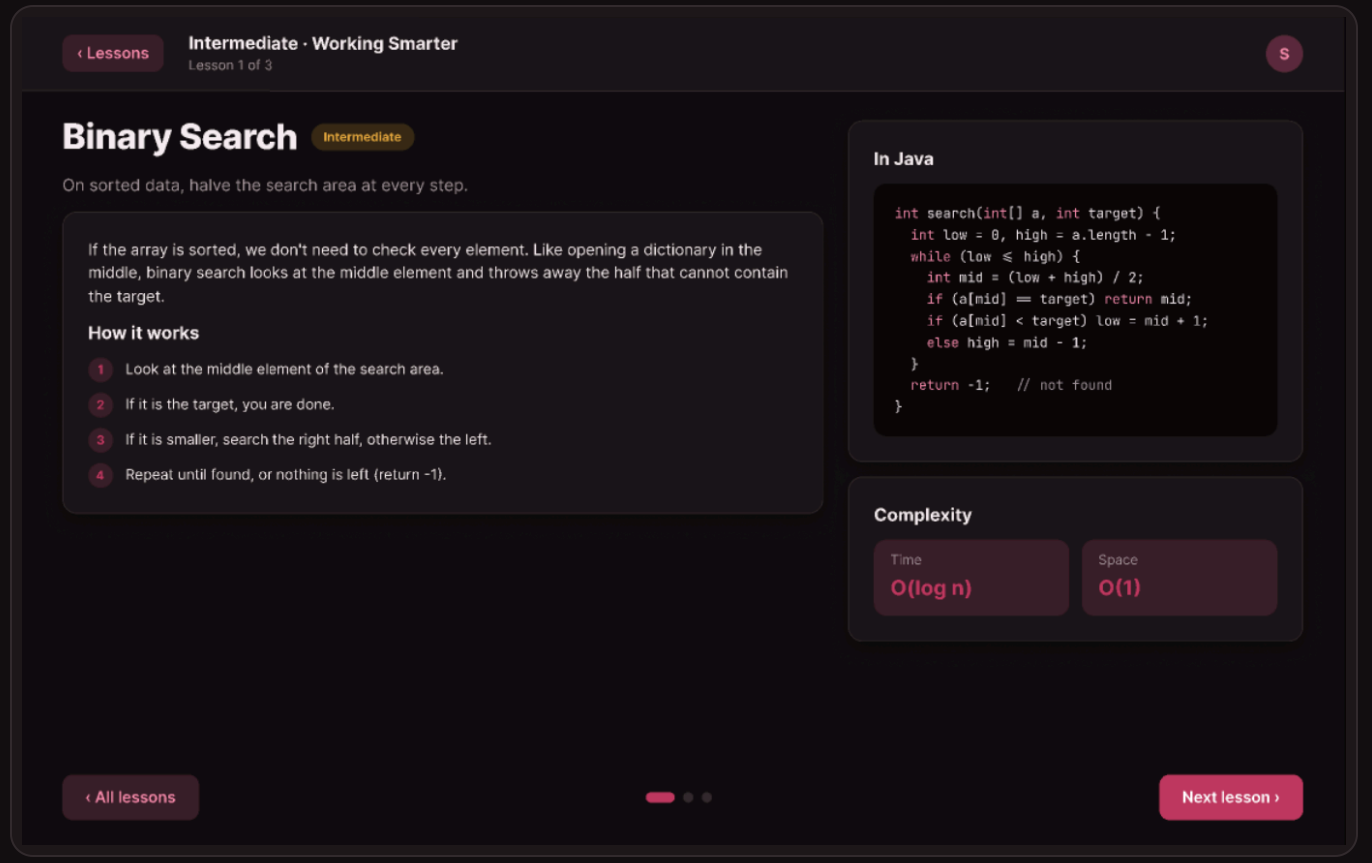
Prototype

Try it: the clickable prototype

Every button leads to the right screen, so the design can be tested like the real app.

1. Home › Start level
2. Binary Search › Next lesson
3. Merge Sort › Start quiz
4. Check answer › Next question
5. Results › Continue

Open the prototype ›



Next steps

Phase 2: building the real app in Java

We turn this design into a working desktop app. Each member builds one part and tests it.

Java 21 · JavaFX · Maven · JUnit tests

Phase 1 — Done

Design, lessons and quiz questions

Phase 2 — Next

Working Java app, shown in our second presentation

T — Toumani

App structure, navigation, home screen and visual style

S — Solaiman

Easy level and the quiz engine

Y — Youssef

Intermediate level and saving progress

F — Feryel

Hard level, graph algorithms and the sorting visualizer

A

Thank you

Questions?

SD Squad

Toumani Adib · Alhamad Solaiman · Youssef Jelassi · Feryel

